

AI Engineer Internship

Credit Risk Intelligence Platform

Neostats AI Engineer Internship Portfolio Project

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Accelerating Credit Decisions with Explainable AI

PROBLEM / CONTEXT

Legacy Decision Bottlenecks

Traditional credit scoring mechanisms often lack transparency and operational speed, which hinders rapid, confident decision-making by financial institutions.

SOLUTION / FUTURE STATE

Explainable Decisioning Platform

An end-to-end AI platform designed to deliver fast, explainable credit decisions. The system integrates a robust Machine Learning predictive model with an intuitive Natural Language-to-SQL chatbot for seamless data interaction. The entire application is packaged within a Dockerized deployment pipeline, ensuring immediate, reproducible environment setup across any infrastructure.

Robust Risk Classification via Advanced Sampling

To address the inherent class imbalance in credit default datasets, the training pipeline utilizes the **Synthetic Minority Over-sampling Technique (SMOTE)**. This ensures the model accurately learns minority class patterns without overfitting.



0.82

ROC-AUC



0.71

F1-Score



0.68

Precision



0.75

Recall

These metrics reflect a highly optimized threshold, successfully maximizing the detection of high-risk profiles (Recall) while maintaining a reliable rate of true positives (Precision).

Comprehensive Analytics & Explainable Risk Scoring

- 01 Exploratory Data Analysis (EDA):** Automatically surfaces 5 critical business insights from historical data.
- 02 ML Risk Scoring:** Categorizes new applicants into clear Low, Medium, or High risk tiers.
- 03 AI Chatbot:** Translates Natural Language to SQL, supporting 5 distinct complex query patterns for non-technical users.
- 04 Explainability:** Integrates SHAP values to provide transparent, feature-level interpretations for every individual prediction.
- 05 Deployment:** Features a 1-command Docker setup for immediate operationalization.

Scalable Architecture Powered by Modern AI Tools

The platform leverages a robust, modern technology stack optimized for machine learning deployment:



Streamlit

Frontend

Built with Streamlit to provide a highly interactive, responsive user interface for data applications.



XGBoost

Modeling

Core predictive capabilities are driven by XGBoost for high-performance gradient boosting.



Claude API

LLM Integration

Intelligent conversational querying and SQL generation are powered by the Claude API.



SQLite

Database

Lightweight, reliable data storage and retrieval managed via SQLite.



Docker Compose

Infrastructure

The entire application is containerized using Docker Compose, guaranteeing consistent, reproducible environments.